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The Impact of Amazon Facilities on Local Economies

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ABSTRACT

A number of large companies deploy facilities—factories, warehouses, shopping malls—that employ thousands of people directly and indirectly. Local governments have been enticing these companies with various financial incentives. But how much economic growth do these facilities promote, if any? We study this question using Amazon distribution facilities as the case in point. We show evidence of a positive effect of opening Amazon's distribution facilities on counties' economic outcomes. We focus on mid-sized counties in which Amazon opened facilities in the years 2014–2017 because we have good controls for these counties. We find a selection effect where Amazon locates its facilities, and we address this issue. Our preferred methodology is Callaway–Sant'Anna difference-in-differences combined with matching. After Amazon's entry, in our preferred specification, we find that the employment-to-population ratio in the treated county increased by 0.0087 (+1.46% at the mean), the poverty rate decreased by 0.36 percentage points (−2.69%), and the median household income increased by \$1413 (+2.33%). We present evidence to argue that our findings can likely be interpreted as causal and plausible.

1 | Introduction

Large companies have an outsized impact on the economy, with much of it coming through direct and indirect employment of a significant workforce. A large automotive plant can easily employ 5,000 people, while a Tesla plant employs more than 10,000. Large retail stores (like Walmart) employ 300–500 people each, and a shopping mall can employ up to 10,000 people. In addition, these large employers often create indirect employment of unknown size through the need for catering, construction, repairs, and schooling. Employing such a significant number of people can have an important impact on the local economy, which is why governments have been enticing large companies with a variety of incentives, and the media covers every new launch extensively. But just how much impact does a large facility have on a local economy? The evidence to date is quite mixed, while governments often offer hundreds of millions of dollars in incentives for each facility.

We use the deployment of Amazon fulfillment centers as a testbed to study the impact of large facilities on county economic outcomes. Specifically, we compare the outcomes in counties with Amazon facilities (“treatment counties”) to those of other counties without Amazon presence (“control counties”). We focus on mid-sized counties for which we have good control, since there are no large counties without Amazon facilities, and small counties rarely have Amazon facilities. We find that in recent years, the treatment counties differ from other counties in the levels and trends of key variables such as income and population age structure. Amazon tends to locate in larger, younger, and faster-growing counties. We address the endogeneity induced by this selection by using nearest-neighbor matching to find control counties that most closely resemble treatment counties, just prior to Amazon's entry. The matching is done on variables that matter the most in site selection, such as population, income, and the fraction of the population in the prime age working group (19–64 years). We estimate county-year panel regressions

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using this “matched-panel” approach. We provide estimates from both the classic two-way fixed effects model and the Callaway–Sant’Anna setup; the latter accounts for any bias due to staggered treatment and heterogeneous effects (Callaway and Sant’Anna 2021; Goodman-Bacon 2021). The pretrends are muted and statistically insignificant in the matched panel specifications. Our preferred specification is the Callaway–Sant’Anna method with the matched panel.

Our final sample consists of 59 mid-sized counties where Amazon was first located between January 1, 2014, and June 30, 2017. The starting year is chosen because Amazon started to grow the network rapidly at that time. We find that it takes about 3 years for the headcount to ramp up after Amazon enters a county. Therefore, we wanted at least 3 years of outcomes after entry. However, COVID-19 restricts our postperiod to no later than 2019. To keep a balanced panel, we focus on average effects over the first 3 years after entry. It should be noted that this is a conservative approach, as the full impact presumably plays out from the third year onward once headcount has fully ramped up. We find that the employment-to-population ratio increases by 0.0087, which corresponds to 1.46% at the mean. The poverty rate decreases by 0.36 percentage points (2.69%). The median household income increased by \$1,413 (2.33%).¹ In general, we conclude that there is statistically and economically significant evidence of a positive economic impact.

Can our findings be interpreted as causal, or are we picking up spurious correlation due to preexisting differential trends in counties that Amazon entered? We perform a number of additional analyses that support a causal interpretation and show that the gain in employment in a county’s warehousing sector is well-matched with the size of the Amazon workforce. In addition, we conduct placebo tests. We would not expect Amazon’s entry to increase county employment in the federal or state government or in manufacturing.² We confirm that this is the case.

1.1 | Literature

This paper is related to the large literature on place-based job policies designed to increase local economic outcomes, particularly employment. There are a variety of such policies that are typically designed to attract firms in the tradable sector.³ The rationale for preferring firms in the tradable sector is straightforward: Incentives to firms in the nontradable sector are unlikely to lead to net job creation. For example, incentives offered to a fast food chain to add jobs will likely reduce revenue and jobs at other nearby fast food outlets. In contrast, incentives to tradable firms can induce a job multiplier effect through supplier effects and worker demand effects. Supplier effects boost employment when local firms expand or new firms enter to supply the tradable sector firm. The effects of worker demand create jobs locally when workers in the tradable sector firm and its suppliers spend on locally produced goods and services. Bartik (2020) presents information on the types and extent of such policies in the United States and reviews evidence on their effectiveness. He estimates that in recent years, place-based job policies amount to about \$60 billion annually. Of this total, the bulk, about \$47 billion, is due to tax credits and cash incentives to individual businesses,

mostly awarded to large establishments in the tradable sector. He suggests that job multipliers range from 1.3 to 1.7 (i.e., for every job created in the tradable sector, an additional 0.3 to 0.7 other local jobs are created). Moretti (2010) investigates long-term job multipliers due to an exogenous increase in employment in the tradable sector, taking into account the endogenous reallocation of factors and price adjustment. He finds multipliers ranging from 2 to 2.5 for every additional trade sector job. Greenstone et al. (2010) demonstrate advantages that counties with new manufacturing plants have over counties without such facilities.

Another related stream of literature studies the impact of large-box retailers on the local economy, including Basker (2005), Goetz and Swaminathan (2006), Neumark et al. (2008), Pope and Pope (2015), and Zheng (2016). Basker (2007) reviews this literature and concludes that there is a small net positive impact on job creation—the median new Walmart store creates about 400 new jobs, but these are largely offset by job losses in competing retailers. The other notable and consistent finding is that Walmart lowers prices for consumers, including those who shop at competing retailers. The consequent decline in profitability drives some of the smaller retailers out of the market. Recently, Zheng (2016) studied the store placement decisions of leading retail chains while considering competitive dynamics and preemption incentives. In addition, Wiltshire (2023) finds that the entry of Walmart supercenters has negative consequences for local earnings and employment. In contrast, we find no impact on local retail jobs in treated counties. Our paper is related to this literature because it studies the local impact of investments, but it also differs in that Amazon not only sells its own products, but also provides marketplace services to third-party sellers, who account for close to 60% of sales.⁴ Importantly, the impact of Amazon facilities might not be as localized as those of Walmart’s since the former serves a much wider geographical footprint; fulfillment centers ship to customers much further away than the usual driving distance between a customer’s residence and a Walmart store. More closely related, Chava et al. (2024) find retail job losses in counties following the construction of warehouses by a major e-commerce retailer. They do not specify the identity of the retailer. However, we note that while our estimates measure effects on total employment, Chava et al. (2024) focus on retail employment in proximate stores. A closely related work is an Economic Policy Institute report from 2018 that investigates net job creation by Amazon facilities (Jones and Zipperer 2018) over a different time period. However, they fail to find evidence of net job creation, whereas we find evidence of job creation, a reduction in the poverty rate, and income growth. In contrast, a recent research paper by the US Chamber of Commerce finds robust employment and income gains within MSAs from the launch of fulfillment centers (Dubai 2022). Finally, very recently Cunningham (2024) shows, using different data and a different level of aggregation, that Amazon has a positive impact on local economies with effect sizes very similar to ours.

This paper also contributes to the broader literature on the changing retail landscape. Hortaçsu and Syverson (2015) review the changes in the retail sector, documenting the rapid growth in e-commerce. They find that this shift is accompanied by gains in labor productivity in the retail sector that are above industry-

TABLE 1 | Data sources.

Data source	Variables
Amazon ^a	Facility monthly headcount, Start dates (inferred)
Bureau of Labor Statistics ^b	Annual county employment-aggregate and by industry ^b
Census	Population-aggregate and by age group ^c
Census - Small Area Poverty and Income Estimates (SAIPE) ^d	Poverty rate, Median household income
US Department of Agriculture - Economic Research Service ^e	Rural-urban continuum county class

Note: Data sources are publicly available from the respective agency websites. ^aProprietary headcount data from Amazon; see main text for a description of how start dates were inferred. ^b<https://www.bls.gov/cew/downloadable-data-files.html>; we averaged across quarters by county, and by county-NAICS as required to get annual estimates. ^cOne-year estimates, 16 years and over, <https://www.census.gov/acs/www/data/data-tables-and-tools/data-profiles/>. ^d<https://www.census.gov/programs-surveys/saie/data/datasets.html>. ^e<https://www.ers.usda.gov/data-products/rural-urban-continuum-codes>.

wide averages. As a consequence, they note that while retail has maintained its share of value-added, it has seen a falling share of labor employment. However, they caution against a naive extrapolation to an online-only future of retail, pointing out that the growth of warehouse clubs and supercenters such as Costco, Sam's Club, and Walmart supercenters has been even more rapid than that of e-commerce. They suggest that projections of an online-only retail sector in the future are simplistic, and instead, a hybrid "bricks-and-clicks" format could play a major role in retail. In fact, Amazon has expanded beyond an online retail format with Amazon Books stores, Whole Foods Market stores, and the more recent advent of Amazon Go convenience stores. Traditional brick-and-mortar retailers increasingly offer services such as "click-and-collect," buy online and pick up or return in store, buy in store and ship to home; they create pop-up store models and other omnichannel retail formats. Here, we are more narrowly concerned with the impact of Amazon's distribution facilities on local economic outcomes, and we do not attempt to measure the nationwide impact of Amazon's growth on retail employment. The latter exercise would necessarily be more speculative, as the retail sector is changing in multiple ways.

The outline of the paper is as follows. We briefly describe the data first, and then we provide a background of Amazon's operations and the location dynamics of its distribution centers, including population and economic trends in relevant counties. Then we present our empirical methodology, followed by the results. Next, we discuss our findings and argue that our results can be interpreted as causal and are also plausible. Finally, we summarize our findings in the conclusion and briefly comment on the limitations of the current study and avenues for future research.

2 | Data

Table 1 lists the data sources for the variables used in the analyses. The start dates of Amazon distribution facilities are inferred from the facility headcount data, which are available to us at the monthly level.⁵ However, we assign a starting year instead of using the starting month as observed in the facility headcount data. We do so because, as shown in Figure 2a,b, headcount at a facility (and at the county level) gradually increases over 3 years. There is no step-function jump in headcount at the start; in the first few months, the headcount is typically low.

We therefore expect the economic impact to begin to appear no earlier than a few months after the start.⁶ We assume that the starting year is t if the facility first appears in these data in the first half of year t , and year $t + 1$ if the facility appears in the second half of year t . We study three economic outcomes—the employment-to-population ratio, the poverty rate, and the median household income, since most Amazon warehouse employees are on the left side of the income distribution, which is better captured by the median. In contrast, outliers on the right side of the distribution would strongly affect the mean. The employment-to-population ratio is the fraction of the adult population (aged 16 or more) that is employed. This labor market indicator has become popular in recent years, as it combines participation in the labor force and unemployment (Donovan 2015).

2.1 | Amazon's Facilities

Amazon's fulfillment network consists of fulfillment centers, sorting centers, specialty fulfillment centers, and a variety of other buildings.⁷ We collectively refer to these buildings as Amazon facilities. Amazon tags each facility with a unique location code. A given facility can be a single building or a cluster of colocated buildings of different types. In July 2020, Amazon operated 532 facilities in 218 counties, many of which opened during the last 2 years. Figure 1a shows the number of new Amazon facilities by year of opening. Figure 1b shows the number of new counties by year of launch of Amazon operations.

Various considerations go into the decision about where to locate Amazon facilities⁸ including proximity to demand, proximity to labor supply, transportation infrastructure, price and availability of real estate, and distribution network requirements, to name a few. We define treatment counties as those with one or more Amazon facilities. Treatment counties are very atypical, with Amazon investing in the more populous regions of the country, driven by the goal of being closer to the maximum number of customers. Table 2 gives a fuller picture of the difference in population and economic variables. Treatment counties differ in both levels and trends. The pattern of selection of counties is consistent with a desire to locate in or around areas of large and growing demand.

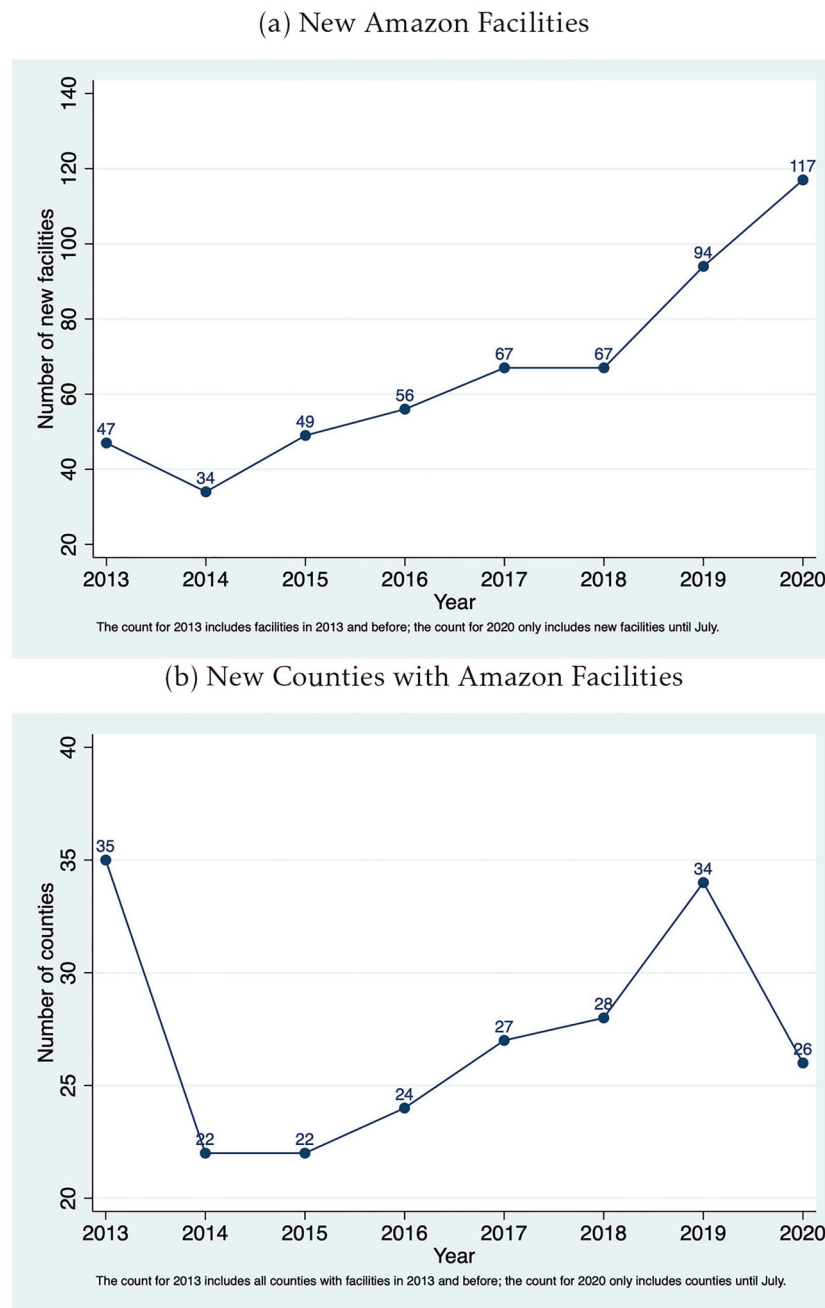


FIGURE 1 | Count of new Amazon facilities and counties. *Note:* Proprietary data showing the yearly count of new Amazon facilities and the number of new counties in which these facilities opened.

There are at least two reasons to expect that Amazon's facilities will have an impact on the local economy. First, as Figure 2a shows, Amazon facilities employ a large number of people. On average, the number of employees at a new facility increases to more than 1000 within the first year of operation. At the county level, on average, the number of employees reaches 3000 at the end of the third year of operation, driven in part by growth within existing facilities and in part by the construction of new facilities (Figure 2b). In the headcount, we include all direct employees, regular or temporary, and full time or part time, as well as third-party and vendor staff working in the facility. In addition, there is unknown but likely substantial indirect employment in logistics,

food sales, maintenance, and so forth. Second, since October 2018, Amazon has raised its starting wage for all US employees to \$15 an hour, a level well above the prevailing minimum wage in many locations, and historically Amazon has paid higher wages compared to many large employers (Wal-Mart, McDonalds, etc.).

3 | Empirical Methodology

We compare outcomes such as employment, income, and poverty rate in treatment counties with those of other counties. We select

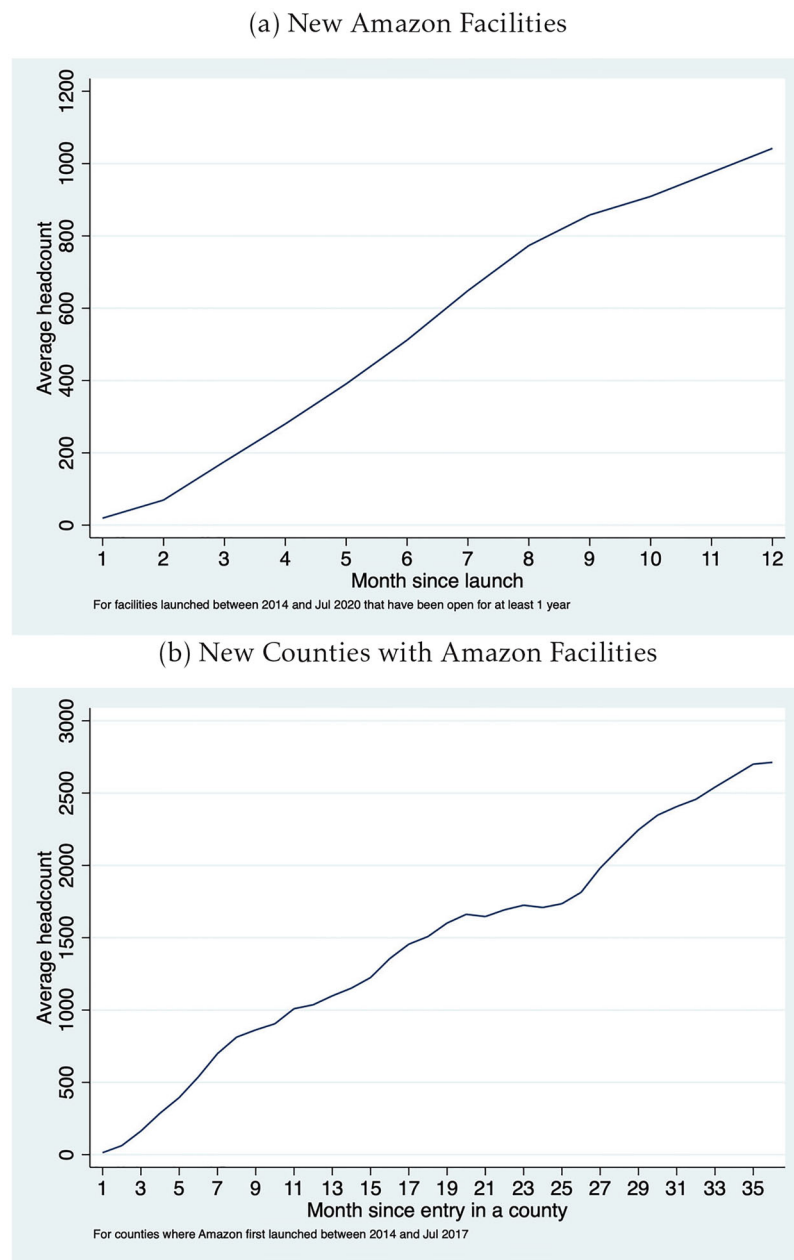


FIGURE 2 | Trend in average headcount at facility and county levels. *Note:* Proprietary Amazon data showing the average headcount over time in a newly opened facility, and aggregated across all facilities in a newly entered county.

the county as the unit of analysis for two reasons. First, reliable annual data are readily available for most counties, excluding some of the most sparsely populated ones. Second, negotiations and planning on the location of the facility often take place at the county level.⁹

We focus on the counties where Amazon was located for the first time after January 1, 2014. Those are the ‘treatment’ counties. We select those dates to ensure that sufficient “before” and “after” data are available. We restrict the pretreatment period to start from 2011 to exclude the years 2008–2010, which were characterized by significant economic disruption due to the financial crisis. Using 2014 as the treatment start date provides 3 years,

that is, 2011, 2012, and 2013, of pretreatment data. We restrict our analysis to the end of 2019, which is the last pre-COVID year; dramatic changes both for Amazon and the economy in general during the pandemic, so we exclude this time period. Therefore, we use June 30, 2017 as the cutoff entry date to ensure that we have at least 3 years of outcome data, that is, 2017, 2018, or 2019, posttreatment, where treatment is defined as Amazon’s entry.

From the pool of “control” or non-Amazon counties, we exclude counties with 2013 populations less than 50,000. Since Amazon has chosen to locate its facilities in the more populous counties (see Table 2), the smaller counties are poor candidates as controls.

TABLE 2 | Population and economic levels and trends: Treatment versus all other counties.

	Population	Empl./Pop. ^a	Median HH income	Poverty rate	Fraction (19–64)
			In 2018		
All other (<i>N</i> = 2923)	51,690 [1555]	0.49 [0.006]	51,506 [234]	15.417 [0.114]	0.565 [0.001]
Treatment (<i>N</i> = 218)	805,497 [66,883]	0.585 [0.011]	70,071 [1,188]	11.755 [0.302]	0.6 [0.002]
			Annual % change (2011–2018)		
All other (<i>N</i> = 2923)	0.006 [0.017]	0.527 [0.032]	2.62 [0.018]	–1.839 [0.035]	
Treatment (<i>N</i> = 218)	0.945 [0.06]	0.982 [0.063]	3.12 [0.065]	–2.811 [0.128]	

Note: Mean values with the standard deviation of the mean in parantheses. ^aEmployment/population NA for counties with population under 65,000.

Other counties that are excluded are those in which Amazon had facilities prior to 2014 and also those where Amazon entered between July 1, 2017, and June 30, 2019. However, we include the most recent treatment counties, that is, those where Amazon first entered after July 1, 2019, since any effects would not show up in the outcomes data before 2020. We also exclude counties in Hawaii and Alaska, as they are somewhat distinct from those in the mainland United States. Finally, we also exclude counties that were adjacent to any Amazon counties as of June 30, 2019, because they could be affected by economic spillovers.

We combine matching with panel data regressions. We do this because treatment counties are very atypical, and by matching, we aim to achieve an apples-to-apples comparison. Specifically, we restrict treatment counties to those that have a population in 2013 between 50,000 and 1,000,000 people. We do so because all counties with more than 1,000,000 people are treated by 2013 and there are no treated counties with fewer than 50,000 people. Our preferred specification is the Callaway–Sant’Anna estimator for the matched panel using the *CSDID* package. However, we also report on the two-way fixed effects estimator for completeness.

3.1 | Constructing the Matched Panel

To construct a matched panel, we first impose the following restrictions: We only include counties with populations in 2013 between 50,000 and 1 million; Amazon was not present in small counties, and there are no large counties without Amazon facilities. We exclude all counties where Amazon had sites before 2014, as well as those counties where Amazon first built sites between July 1, 2017, and June 30, 2019. However, note that we allow counties that Amazon entered after June 30, 2019, to serve as potential matches.¹⁰ The final set consists of 59 treatment counties (those that received their first Amazon facility between January 1, 2014, and June 30, 2017).

We apply the same restrictions as listed above to the control counties with the additional constraint that the population in 2013 should be less than 1 million. That yields 442 potential control counties. See Appendix Table A1 for the list of treatment counties along with their population in 2013 and the start date of their first Amazon facility. We use nearest-neighbor matching to find the closest counterpart from the potential controls for each treatment county in 2013, that is, just prior to the start of treatment.¹¹ The matching algorithm picks county *j* as a match for county *i* by minimizing the distance $(x_i' V x_j)^{\frac{1}{2}}$ where *x* is the vector of matching variables, and *V* is the diagonal weighting matrix comprising the inverse of the variances of each element of *x*.

We pick the matching variables to be those that are statistically significant in a cross-county probit regression of county-level covariates. The population and median household income in 2013 are highly significant, and we try to match these two variables as accurately as possible. In addition, we also match the fraction of population of prime working age (19–64 years) in 2013, as well as the change in population from 2011 to 2013. The matching algorithm selects 39 control counties, with some of those being matched to more than one treatment counties. In the matched panel regressions, we use frequency weights to reflect the frequency with which control counties are picked as matches. See Appendix Table A2 for the list of matched counties along with their population in 2013, and frequency weights. Note that in a number of cases, Amazon enters the county after mid-2019. In other words, some of the most recent treatment counties are good matches for the earlier ones.

Table 3 displays the mean and standard errors for key population and economic variables in 2013 for the treatment counties and the matches. There is a far better balance in the levels and trends of the key variables between the two sets of counties compared to what we saw in Table 2. Even though we have a matched panel, we still control for county-type linear trends in case there is residual selection. The key identification assumption is that the

TABLE 3 | Covariate balance between treatment and matched midsized counties.

	Population	Empl/Pop.	In 2013 Median HH income	Poverty rate	Fraction (19–64)
Amazon	547,325	0.594	60,427	13.275	0.61
(<i>N</i> = 59)	[37,327]	[0.019]	[1626]	[0.623]	[0.004]
Matched	520,302	0.555	60,143	13.53	0.602
(<i>N</i> = 39)	[36130]	[0.018]	[1702]	[0.626]	[0.003]
<i>p</i> -Value	0.604	0.145	0.904	0.771	0.065
	% change from 2011 to 2013				
Amazon	1.892	1.148	4.853	−1.119	
(<i>N</i> = 59)	[0.223]	[0.316]	[0.451]	[1.25]	
Matched	1.789	1.257	4.298	−0.986	
(<i>N</i> = 39)	[0.211]	[0.311]	[0.529]	[0.892]	
<i>p</i> -Value	0.737	0.807	0.427	0.94	

Note: ^aThe 39 matched counties are weighted to reflect the frequency in match. If a county was a match to *m* treatment counties, it has a weight of *m*. The weights add to 59. ^bFrom a two-tailed *t* test of difference in means.

treatment and control counties have common preexisting trends in outcomes, which we demonstrate shortly.

3.2 | Matched Panel Regressions

We run panel regressions using annual county data from 2011 to 2019:

$$y_{it} = \alpha_i + \alpha_t + \theta_c * t + \beta Amz_i * Post_{it} + \epsilon_{it}. \quad (1)$$

Here y_{it} is the outcome in county *i* in year *t*. Treatment counties are denoted by *Amz*—these are 59 counties where Amazon was first located between January 1, 2014 and June 30, 2017. The post period is assumed to begin in year *t* if Amazon is located first in a county in the first half of year *t* and in year *t* + 1 if it was the second half of year *t*. The coefficient β captures the effect of interest. The α 's are fixed effects for the county and year. We also allow for different time trends for different types of counties. County type is one of the nine classes in the 2013 Rural–Urban continuum code that distinguishes metropolitan counties by the population size of their metro area, and nonmetropolitan counties by degree of urbanization and adjacency to a metro area.¹² Different types of counties could have different economic trends. Since Amazon fulfillment centers are almost always located in populous metropolitan/urban counties, it is important to control for differential trends to avoid picking up spurious correlation. Later, we illustrate the role of selection by showing how β changes as we progressively add more controls.

Identification in this method relies on the separate state and county-type time trends absorbing all preexisting differential trends in outcomes, so that β captures deviations from those trends induced by Amazon's entry.

We also estimate the following event study specification using STATA's *eventdd* package:

$$y_{it} = \alpha_i + \alpha_t + \theta_c * t + \sum_{j=-2}^{-4} \beta_j (Lead_j)_{it} + \sum_{k=0}^3 \gamma_k (Lag_k)_{it} + \epsilon_{it}. \quad (2)$$

The lags (leads) are dummy variables that indicate a county was a certain number of periods away from the event, that is, Amazon entry at time “0.” The lead of −4 accumulates all years prior to and including 4 years before the event. The final lag of +3 accumulates all years after and including 3 years after the event. The coefficients for lags 0 to 2 are for a balanced panel, as all treated counties have at least 3 years of data in the sample postentry—the counties where Amazon first locates in 2017 have the smallest post window of 3 years as the data end in 2019.

Our regression specifications are two-way fixed-effects models, that is, difference-in-differences models with county and time fixed effects. A recent literature in econometrics has shown that such models yield a weighted average of all possible permutations of pairwise difference-in-differences estimators, where a pair is the never treated control group paired with a cohort of counties treated at time *t*, or a cohort of counties treated at time *t* paired with another cohort of counties treated at time $t^* > t$ (Callaway and Sant'Anna 2021; Goodman-Bacon 2021). Our preferred estimates are based on comparisons of treated counties with the never treated control group using *CSDiD* STATA package developed by Rios-Avila et al. (2022).

4 | Results

Here, we present the findings for the 59 midsized treatment counties. For completeness, we also present the full panel TWFE and CSDiD regressions in Appendix Tables A4 and A5, and document the strong selection in the choice of counties to

TABLE 4 | Results for 59 midsized treated counties without matching.

	Empl./Pop.	Poverty rate	Median HH income
Amz*Post, levels	0.010*** [0.003]	−0.443*** [0.194]	2157.477** [566.37]
Mean value, levels	0.594	13.274	60427.31
Obs.	3999	5099	5099
R ²	0.992	0.942	0.975

Note: Robust standard errors clustered on county in parentheses. Regressions include county and year fixed effects, and county-type specific linear time trends. Only include treated counties with 2013 population >50,000 & <1,000,000. The number of treatment counties is 59: those that have start dates between January 1, 2014 and June 30, 2017. As control counties, we only include counties that are not adjacent to Amazon counties as of July 2019, and have 2013 population >50,000 for a total of 508 control counties. Employment–population ratio is not available for counties with population <65,000. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE 5 | Matched panel results for 59 midsized treated counties.

	Empl./Pop.	Poverty rate	Median HH Income
Amz*Post, levels	0.007* (0.003)	−0.429** (0.214)	1190.892* (616.373)
Mean value, levels	0.594	13.274	60,427.31
Obs.	1047	1062	1062
R ²	0.993	0.961	0.974

Note: Robust standard errors clustered on county in parentheses. There are 59 treatment counties and 39 matched counties. If a control county matched to m treatment counties, it had a weight of m in the regressions. Regressions include county and year fixed effects. Employment–population ratio not available for counties with population <65,000. Mean values are for the 59 treated counties. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE 6 | ATT estimates for matched panel from Callaway–Sant’Anna DiD.

	Empl./Pop.	Poverty rate	Median HH income
Matched panel, levels	0.0087*** (0.002)	−0.358* (0.19)	1413.05*** (494.553)
Mean value, levels	0.594	13.274	60,427.31
Percentage change, %	1.46	−2.69	2.33

Note: Robust standard errors clustered on county in parentheses. Average treatment effect on treated estimates from the *CSDID* package in STATA restricting comparisons to never-treated controls only. There are 59 treatment counties and 39 matched counties in the matched panel. If a control county matched to m treatment counties, it had a weight of m in the regressions. Regressions include county and year fixed effects. Employment–population ratio not available for counties with population <65,000. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

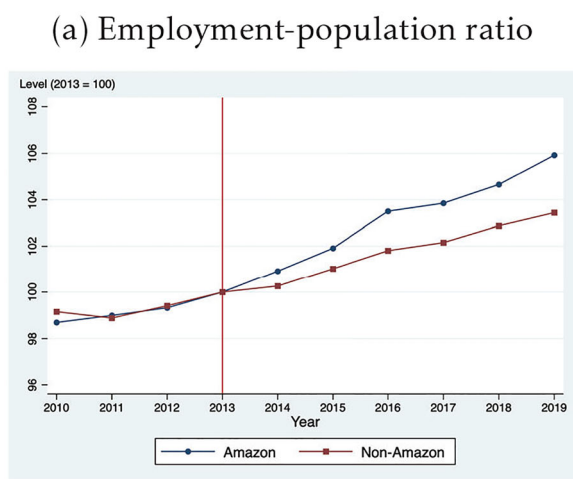
locate Amazon facilities. Table 4 reports the results of the panel regression with the 59 counties, first without matching.

Figure 3a–c shows the trend in the three outcomes for the treated and control counties in the matched panel. Although there are pretrends in employment–population ratio and in median household income, there do not appear to be differential trends for treatment versus control counties. There are no pretrends for either set of counties in the case of poverty rate.¹³

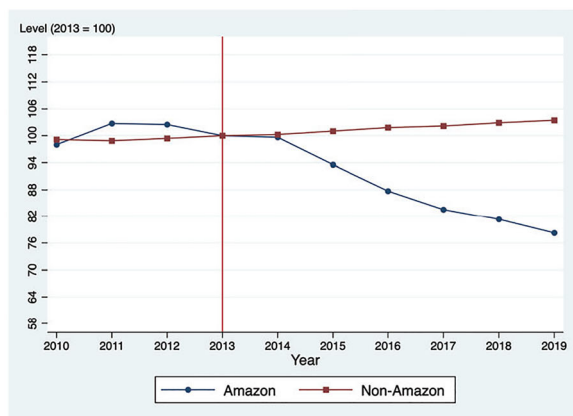
Table 5 reports the results of the matched panel regressions. There is a 1.18% increase in the employment/population ratio, a 3.12% decrease in the poverty ratio, and an increase of 1.97% in the median household income. If the matching process successfully removes pretrends, the matched panel regressions should find diminished effect sizes for both the employment–population

ratio and median household income, both of which exhibited distinct pretrends in the full panel analysis. That is indeed the case (compare with Appendix Table A4). The effect sizes for employment–population ratio and median household income are indeed diminished. For the poverty rate that had no pretrend, the effect size is similar in the full and matched panel.

Figure 4a–c shows the event study estimates for the matched panel. The event timeline is laid out, with 0 being the year when Amazon first locates in a county. Since the last cohort of treated counties saw Amazon locating in 2017, and the data run up to 2019, the effects for years 0 to 2 are estimated in a balanced panel of all 59 treated counties. The year 3 estimates are multiplied by all subsequent lagged effects, but those estimates represent a mix of lags and a changing composition of treated counties. Of note, compared to the full panel, the muting of differential pretrends



(b) Poverty rate



(c) Median household income

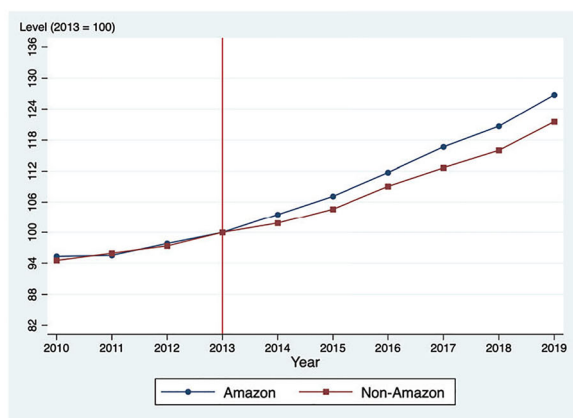
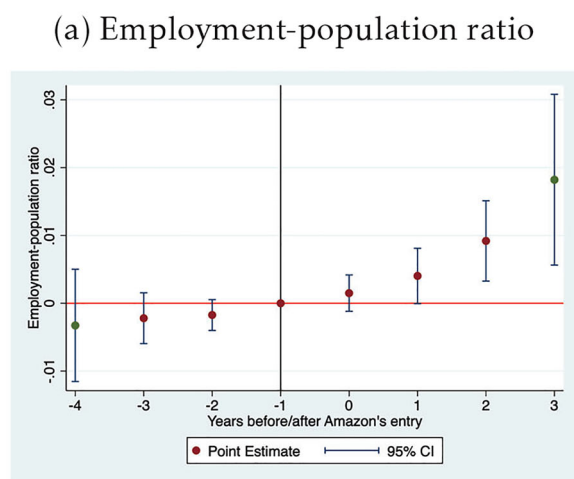
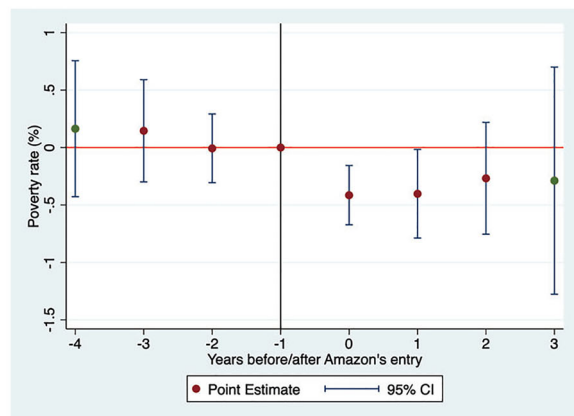


FIGURE 3 | Trends in outcomes in matched panel counties. *Note:* Time trend in outcomes for 59 treated counties and 39 matched control counties for 10 years for a total of 980 observations. The outcome variables in panels (a)–(c) are Employment–population ratio (fraction), Poverty Rate (%), and Median Household Income (\$), all normalized to base 100 in year 2013.

is clearly visible. None of the lead coefficients are significant, although the point estimates suggest mild differential pretrends with the treated counties on a slightly better trajectory in the preperiod. The lagged coefficients show statistically significant



(b) Poverty rate



(c) Median household income

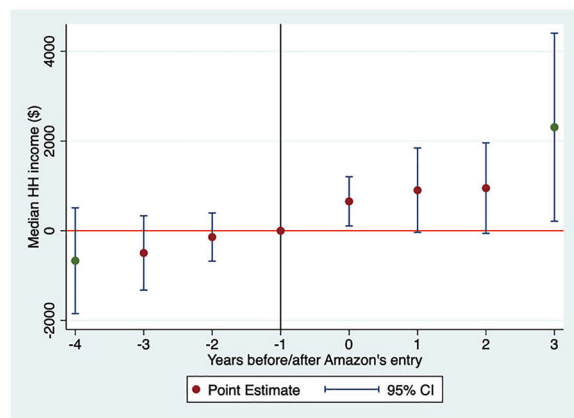
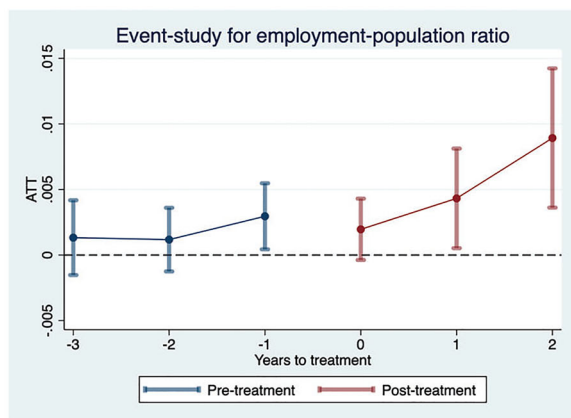


FIGURE 4 | Event study for matched panel. *Note:* Event study graphs from TWFE regressions for the matched panel which has 59 treated counties and 39 matched control counties for 10 years for a total of 980 observations. The outcome variables in panels (a)–(c) are Employment–population ratio (fraction), Poverty Rate (%), and Median Household Income (\$), all in levels.

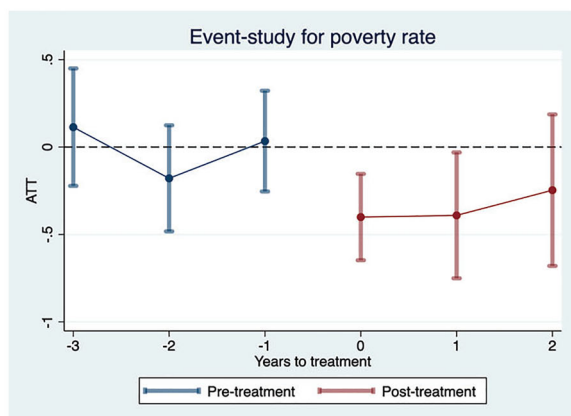
(marginal in the case of poverty rate) and positive effects for all three outcomes.

Figure 5a–c graphs the Callaway–Sant’Anna estimates for the matched panel. As is evident, the pretrends for employment–population ratio and median household income are now muted—

(a) Employment-population ratio



(b) Poverty rate



(c) Median household income

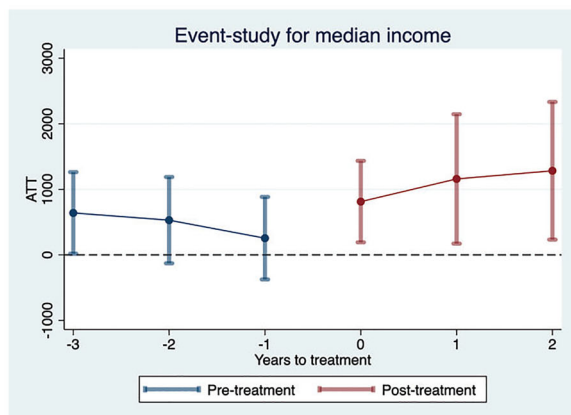


FIGURE 5 | Callaway–Sant’Anna event study for matched panel. *Note:* Event study graphs from Callaway–Sant’Anna difference-in-differences regressions for the matched panel which has 59 treated counties and 39 matched control counties for 10 years for a total of 980 observations. The outcome variables in panels (a)–(c) are Employment–population ratio (fraction), Poverty Rate (%), and Median Household Income (\$), all in levels.

all but one of the lead coefficients are now statistically insignificant. As always, there is no pretrend for poverty rate. The lagged coefficients show positive economic effects. The second row of Table 6 displays the ATT effects for the matched panel. The

TABLE 7 | Effects on employment in different sectors.

	Matched Panel			
	log (trade) ^a	log (govt.) ^b	log (mfg.) ^c	log (retail) ^d
Amz*Post	0.046*** (0.016)	0.055 (0.041)	0.018 (0.017)	0.021 [0.019]
Observations	1062	1062	1057	1062
R ²	0.995	0.975	0.995	0.990

Note: Robust standard errors clustered on county in parentheses. ^aTrade, Transportation, and Utilities. ^bFederal and state government. ^cManufacturing employment. ^dRetail trade. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

employment–population ratio increases by 1.46%, the poverty rate decreases by 2.69%, and the median household income increases by 2.33%. All effects are statistically significant.¹⁴

5 | Discussion

We present additional evidence that our findings can be interpreted as the causal effect of Amazon’s distribution facilities. Table 7 shows evidence to support a causal interpretation of the findings, by measuring the effects on employment in different sectors. A sanity check is to measure the impact on employment in the Trade, Transportation, and Utilities sector, including warehouses. There is a significant and large effect of about 5% in the matched panel that is mapped to about 1600 additional jobs in that sector based on a mean pretreatment employment of approximately 32,000 in a county. This matches well with the average county-level treatment headcount of 2000 over the first 3 years. Columns 3 and 4 show the results of the placebo tests. We should not find any effect on government employment (federal and state combined) or on manufacturing employment. As columns 3 and 4 show, we do not find significant effects on these sectors.¹⁵ We also report the estimated effect on retail employment in column 5—we find no impact. This is in contrast to Chava et al. (2024), who find retail job losses in proximate stores after the construction of warehouses by a major e-Commerce retailer. Instead, we compare retail employment in counties with and without Amazon’s presence using different methods.

The reader wishing to replicate the results of our paper may find it hard without access to Amazon data. Fortunately, public data allow for similar analysis, which is consistent with our results so far. We repeated an analysis using the Callaway–Sant’Anna DiD approach on publicly available data from the MWPVL Supply Chain Consulting company <https://www.mwpvl.com/>.¹⁶ Table A7 in the Appendix displays the summary statistics for Amazon and the matched control counties using MWPVL data. Table A8 in the Appendix displays the ATT estimates for the matched panel approach using Amazon data (our proprietary) and publicly available MWPVL data. In Appendix Figure A3a,b,c, we provide visualizations of the estimates of the event study for the employment–population ratio, the poverty rate, and the median income, respectively. We note that effect sizes and significance are comparable or better using MWPVL data than our data.

To illustrate how economically meaningful our findings are, we use the estimate for the employment-to-population ratio of 0.87 p.p. We apply that to the 16+ population of a typical midsized county (the denominator of the employment-to-population ratio). Taking the midpoint of the population of those counties (525,000) and the share which is 16+ (85%), we get 446,000 as the 16+ population. A 0.87 p.p increase implies 3900 more employed people. Recall that the typical Amazon county adds about 3000 new jobs by the end of year 3. Our estimates are the average over the first 3 years, and so we would expect about 2000 new direct jobs. Hence, there are an additional 1900 indirect jobs or a multiplier close to 2. This finding compares well with the results of Moretti (2010), who finds multipliers of 1.6–2.5. We caveat that some of the effects for later opened warehouses in later years could be partially due to Amazon raising its starting wage for all US employees to at least \$15 an hour. At the same time, it helps to know that Amazon adjusts wages all the time: many facilities were already paying higher than this wage before the announcement.

6 | Conclusion

We analyze the impact of large facilities that large corporations establish in the United States, using Amazon distribution facilities as a test bed. We find a significant positive impact on the economies of the host counties using a variety of methods. Our sample consists of 59 counties where Amazon first built a facility between 2014 and mid-2017. The challenge of causal identification is due to the selective location of facilities; Amazon tended to place its facilities in faster growing and more economically dynamic counties. Our preferred estimates come from Callaway–Sant’Anna difference-in-difference combined with matching. For two reasons, we emphasize the average effects over the first 3 years after entry as well as just the effect in year 3. First, we have a balanced panel of counties in the first 3 years after entry. Second, there is a steady ramping up of headcount up to the third year (possibly due to network expansion with nearby facilities in the same county), so that we expect to see close-to-full effects becoming visible only by that year. We find, in our preferred specification, that the employment-to-population ratio increases by 0.0087, an increase of 1.46% at the mean, the poverty rate decreases by 0.36 percentage points (–2.69%), and the median household income increases by \$1,413 (+2.33%). In general, we conclude that there is evidence of a positive and significant economic impact.

We believe that our results will be of interest to academics and policymakers for a number of reasons. First, several studies have been published on Walmart, another large employer in the retail sector, reporting effects that were small or nonexistent, while positive effects for large manufacturing plants are well-documented. We find large and statistically significant effects for the employment–population ratio, poverty, and median household income. Second, there is considerable ongoing public policy debate about the return on local tax breaks and other fiscal incentives provided to large firms. Therefore, our study provides timely and rigorous empirical evidence of the local economic impact of Amazon.

Our study has a number of opportunities for future research. First, it would be instructive to unpack the mechanisms underlying our findings. For instance, how much of the employment effect is working through the supply side—the direct job gains due to Amazon as well as other local firms expanding employment to provide services to Amazon facilities? And how much is stemming from the demand side—workers employed at Amazon spending their income on goods and services within the county? Second, we focus on the effects of Amazon’s entry in the counties where Amazon opened facilities, but we ignore adjacent counties where Amazon employees could reside or spend money. Future work can study spillover effects in adjacent counties. Third, most of our sample period precedes the big jump in Amazon’s starting hourly wage in 2018 when it was increased to \$15. As more recent data become available, it would be useful to investigate whether that increase in wages had effects beyond the opening of distribution facilities.

Acknowledgments

Our heartfelt thanks to Nico Melton for outstanding research assistance. The authors acknowledge that they are employed by Amazon.com, which is the focus of the analysis in this paper.

Data Availability Statement

The data that support the findings of this study are openly available in Harvard Dataverse, at <https://doi.org/10.7910/DVN/THID0C>. However, some of the data used for this project are confidential and were provided by Amazon under conditions of nondisclosure that preclude us from making those data publicly available. Please see Section 2 for more details. Researchers interested in accessing those data should contact Adam Sedo, adamsedo@amazon.com.

Endnotes

¹These outcome variables are measured in the following units: employment-to-population ratio is a fraction between 0 and 1, poverty rate is a percentage between 0 and 100, and median household income is dollars. We will consistently report the impact as a percentage change over the baseline value of these outcomes

²While we do not detect a statistically significant effect in the initial years, it is plausible that, over time, manufacturers might choose to locate close to Amazon warehouses to minimize the cost of shipping to the fulfillment network.

³The tradable sector produces goods or services that can be traded across regions and international borders. In contrast, the nontradable sector comprises production activities that cannot be easily imported or exported across regions, for example, health care, dining out, and so forth.

⁴<https://www.aboutamazon.com/news/company-news/2020-letter-to-shareholders>.

⁵Interested researchers can also access readily available data on Amazon facility location, headcount, and opening dates here: https://www.mwpl.com/html/amazon_com.html, and we provide supplementary analysis using these data in the Appendix, or by scraping news wires on Amazon warehouse openings. The reader should note that the start date inferred from headcount data would precede the launch of operations in a facility, as it would take a while to recruit and train a critical mass of workers before operations can begin.

⁶The launch of a facility can span years, starting with the signing of the lease, the refurbishment of an existing building, or the construction of

a new one, and the recruitment of workers. There could be a positive economic impact from capital expenditure in the run-up to the start of operations.

⁷See <https://www.aboutamazon.com/workplace/facilities>.

⁸See Houde et al. (2017) for a detailed discussion of the factors that guided the location of Amazon facilities over time.

⁹We have also attempted analysis at zip code level and tract level, but outcome data at those levels are often unavailable, and where it is available, it can be very noisy.

¹⁰We assume that any effects on such counties in the 2019 data would be negligible.

¹¹We use the STATA package *nnmatch* to pick the nearest neighbor to each treatment county.

¹²<https://www.ers.usda.gov/data-products/rural-urban-continuum-codes/documentation/>. Appendix Table A3 displays the distribution of all counties and treatment counties by the rural–urban continuum code. It is clear that Amazon is located almost exclusively in the metropolitan counties.

¹³A caveat is in order—the plots are not in event time and do not capture pretrends for counties where Amazon first located after 2014.

¹⁴In Appendix Table A6, we present the full set of Callaway–Sant’Anna estimates by event year spanning –3 to +2 in event time with 0 being the year of Amazon entry.

¹⁵The same pattern also holds for the full panel, with a significant positive impact on employment in the Trade, Transportation, and Utilities sector, and no effects on employment in the government, manufacturing, or retail sectors.

¹⁶We would like to thank the editor and one anonymous referee for this suggestion.

References

- Bartik, T. J. 2020. “Using Place-Based Jobs Policies to Help Distressed Communities.” *Journal of Economic Perspectives* 34, no. 3: 99–127.
- Basker, E. 2005. “Job Creation or Destruction? Labor Market Effects of Wal-Mart Expansion.” *Review of Economics and Statistics* 87, no. 1: 174–183.
- Basker, E. 2007. “The Causes and Consequences of Wal-Mart’s Growth.” *Journal of Economic Perspectives* 21, no. 3: 177–198.
- Callaway, B., and P. H. Sant’Anna. 2021. “Difference-in-Differences With Multiple Time Periods.” *Journal of Econometrics* 225, no. 2: 200–230.
- Chava, S., A. Oettl, M. Singh, and L. Zeng. 2024. “Creative Destruction? Impact of E-Commerce on the Retail Sector.” *Management Science* 70, no. 4: 2168–2187.
- Cunningham, E. R. 2024. “Local Labor Market Effects of Amazon.” Working Paper.
- Donovan, S. A. 2015. “An Overview of the Employment–Population Ratio.” Congressional Research Service 7-5700.
- Dubai, C. 2022. “The Community Economic Impact of New Distribution Centers.” Tech. rep., U.S. Chamber of Commerce.
- Goetz, S. J., and H. Swaminathan. 2006. “Wal-Mart and County-Wide Poverty.” *Social Science Quarterly* 87, no. 2: 211–226.
- Goodman-Bacon, A. 2021. “Difference-in-Differences With Variation in Treatment Timing.” *Journal of Econometrics* 225, no. 2: 254–277.
- Greenstone, M., R. Hornbeck, and E. Moretti. 2010. “Identifying Agglomeration Spillovers: Evidence From Winners and Losers of Large Plant Openings.” *Journal of Political Economy* 118, no. 3: 536–598.
- Hortaçsu, A., and C. Syverson. 2015. “The Ongoing Evolution of US Retail: A Format Tug-of-War.” *Journal of Economic Perspectives* 29, no. 4: 89–112.

Houde, J.-F., P. Newberry, and K. Seim. 2017. “Economies of Density in E-Commerce: A Study of Amazon’s Fulfillment Center Network.” Tech. Rep. w23361, National Bureau of Economic Research.

Jones, J., and B. Zipperer. 2018. “Unfulfilled Promises: Amazon Fulfillment Centers Do Not Generate Broad-Based Employment Growth.” Tech. Rep. 138921, Economic Policy Institute.

Moretti, E. 2010. “Local Multipliers.” *American Economic Review* 100, no. 2: 373–377.

Neumark, D., J. Zhang, and S. Ciccarella. 2008. “The Effects of Wal-Mart on Local Labor Markets.” *Journal of Urban Economics* 63, no. 2: 405–430.

Pope, D. G., and J. C. Pope. 2015. “When Walmart Comes to Town: Always Low Housing Prices? Always?” *Journal of Urban Economics* 87: 1–13.

Rios-Avila, F., P. Sant’Anna, and B. Callaway. 2022. “CSDiD: Stata Module for the Estimation of Difference-in-Difference Models With Multiple Time Periods.” Working Paper.

Wiltshire, J. C. 2023. “Walmart Supercenters and Monopsony Power: How a Large, Low-Wage Employer Impacts Local Labor Markets.” Working Paper.

Zheng, F. 2016. “Spatial Competition and Preemptive Entry in the Discount Retail Industry.” Mimeo, Available at SSRN: <https://ssrn.com/abstract=2775498>

Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Figure A1: Event Study for Full Panel **Table A1:** List of Treatment Counties in the Matched Panel **Table A2:** List of Matched Counties **Table A3:** Number of counties by Rural–Urban Continuum Code (2013) **Table A4:** TWFE Results From Full County-Year Panel **Table A5:** ATT Estimates for Full Panel from Callaway–Sant’Anna DiD **Table A6:** Callaway–Sant’Anna Estimates by Event Year for Matched Panel **Table A7:** Summary Statistics for matched panel using MWPVL data **Table A8:** CSDiD results for matched panel: Amazon vs MWPVL data **Figure A2:** Callaway–Sant’Anna Event Study for Full Panel **Figure A3:** Callaway–Sant’Anna Event Study for MWPVL